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Goldenberg et al.

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(54) **SLIM COMPACT LENS OPTICAL IMAGE STABILIZATION**

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H04N 23/68 (2023.01)
G03B 5/04 (2021.01)
G03B 30/00 (2021.01)

(52) **U.S. Cl.**
CPC **H04N 23/687** (2023.01); **G03B 5/04** (2013.01); **G03B 30/00** (2021.01); **G03B 2205/0015** (2013.01)

(58) **Field of Classification Search**

CPC H04N 23/687; G03B 5/04; G03B 30/00; G03B 2205/0015

See application file for complete search history.

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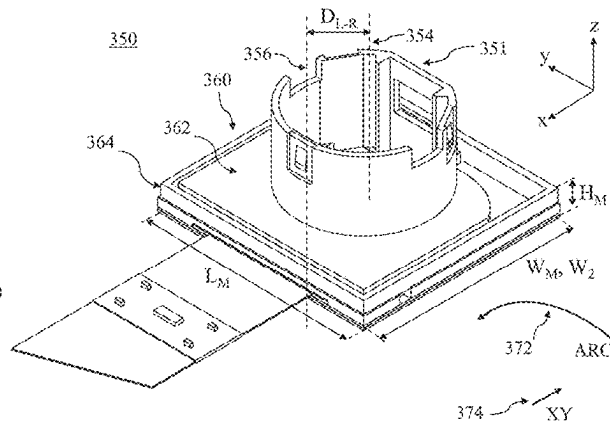
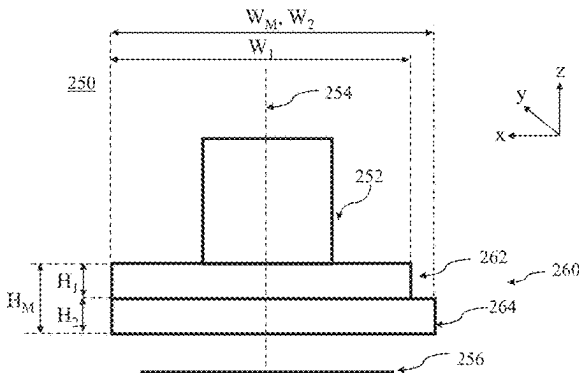
Primary Examiner — Padma Haliyur

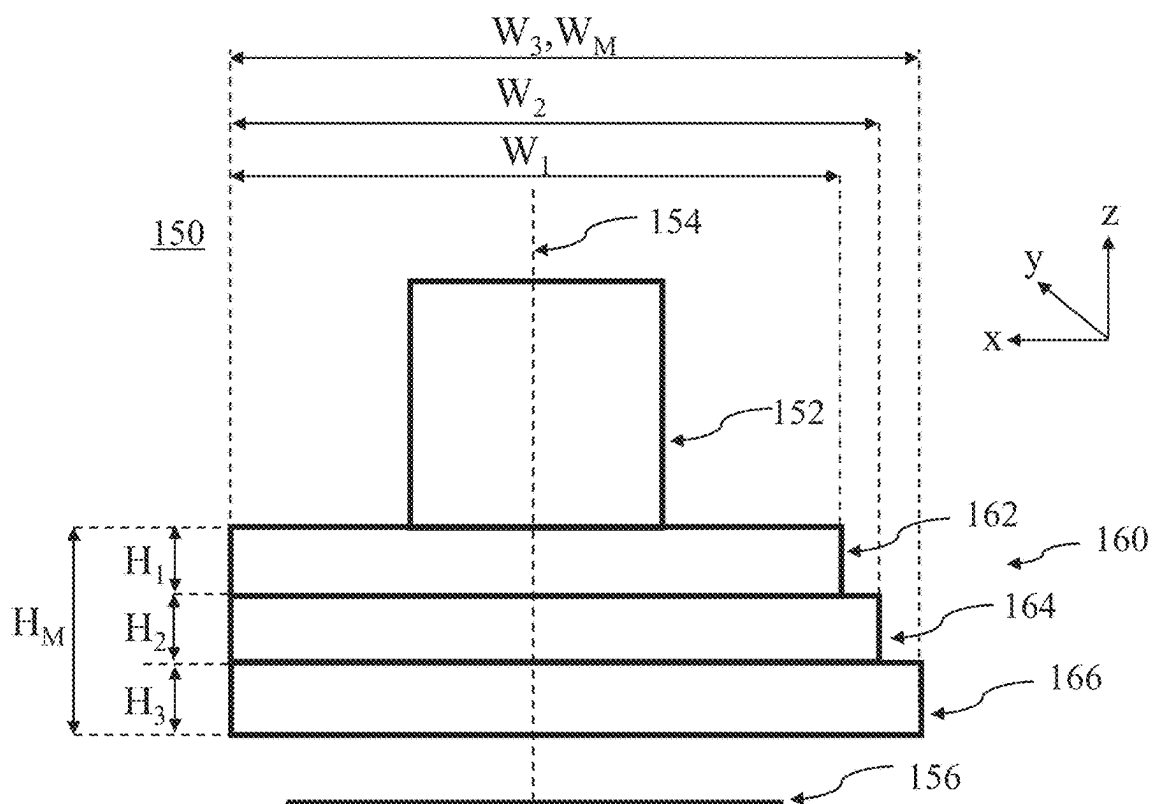
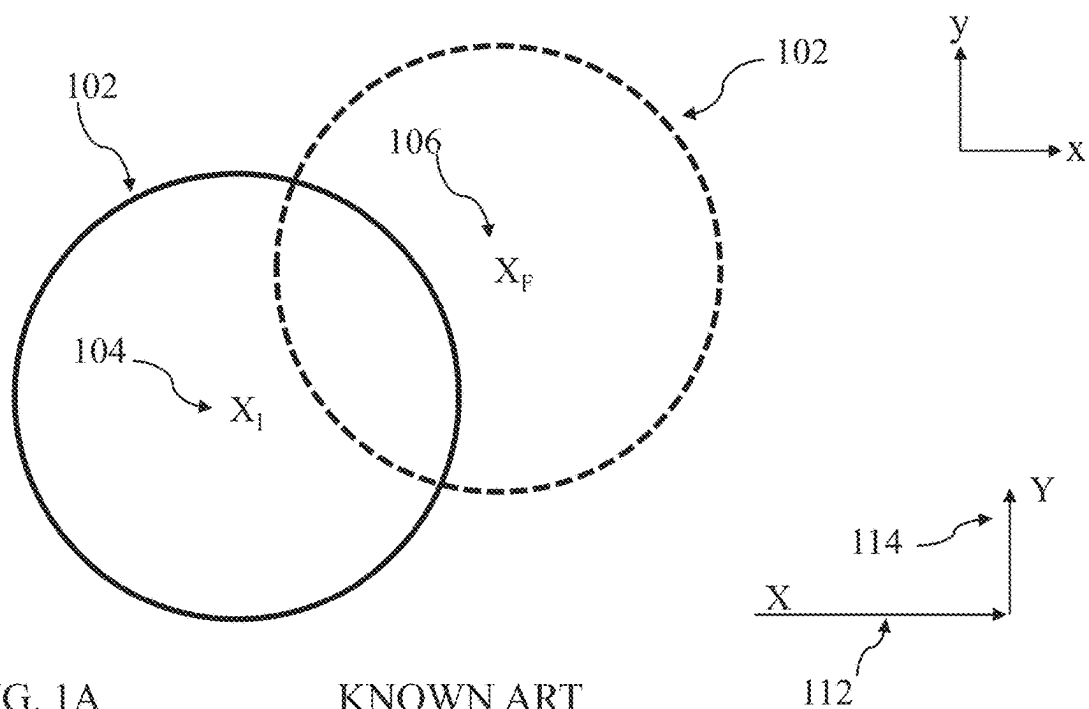
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(57) **ABSTRACT**

Optical image stabilization (OIS) mechanisms, comprising a moving frame including a first groove, a static frame including a second groove, an OIS actuator, and first, second and third bearings that define an OIS plane, wherein the first ball bearing is located in a rail formed by the first groove and the second groove, wherein the ball-bearings are positioned between the moving frame and the static frame and allow a first movement and a second movement of the moving frame relative to the static frame, wherein the first movement is a rotational movement performed around a rotation axis that coincides with the position of the first ball bearing and is perpendicular to the OIS plane, and wherein the second movement is a linear movement along the rail.

20 Claims, 7 Drawing Sheets





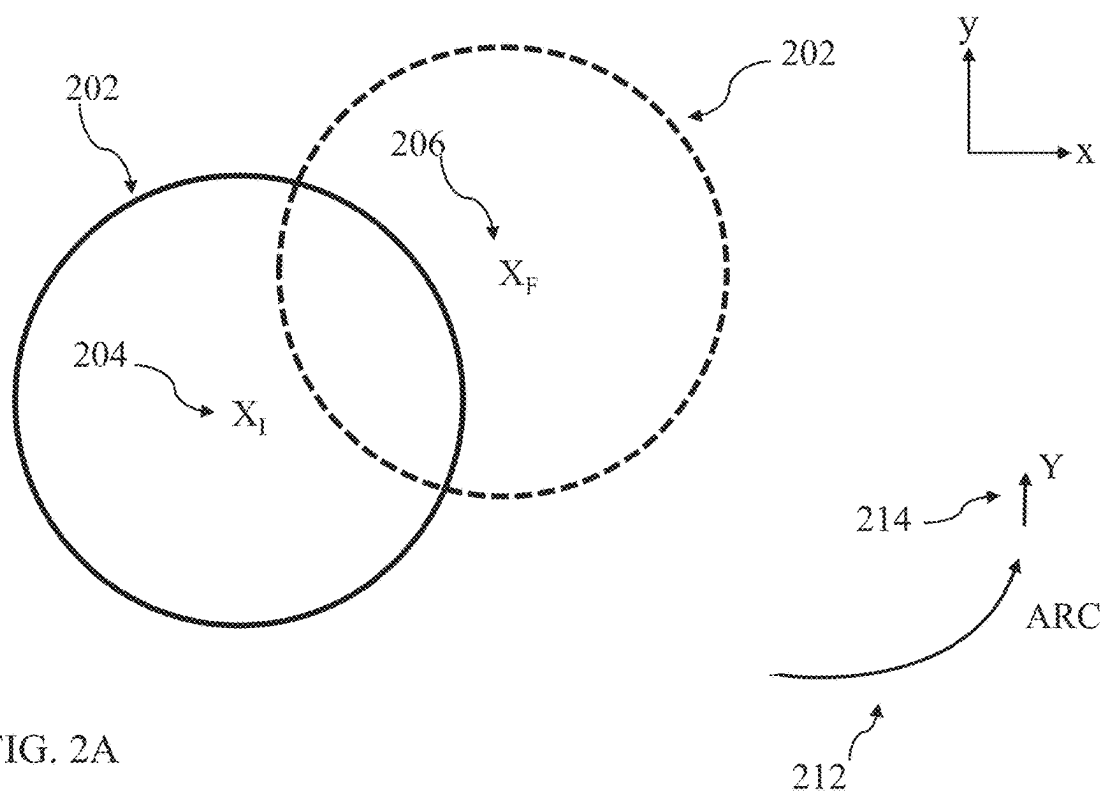


FIG. 2A

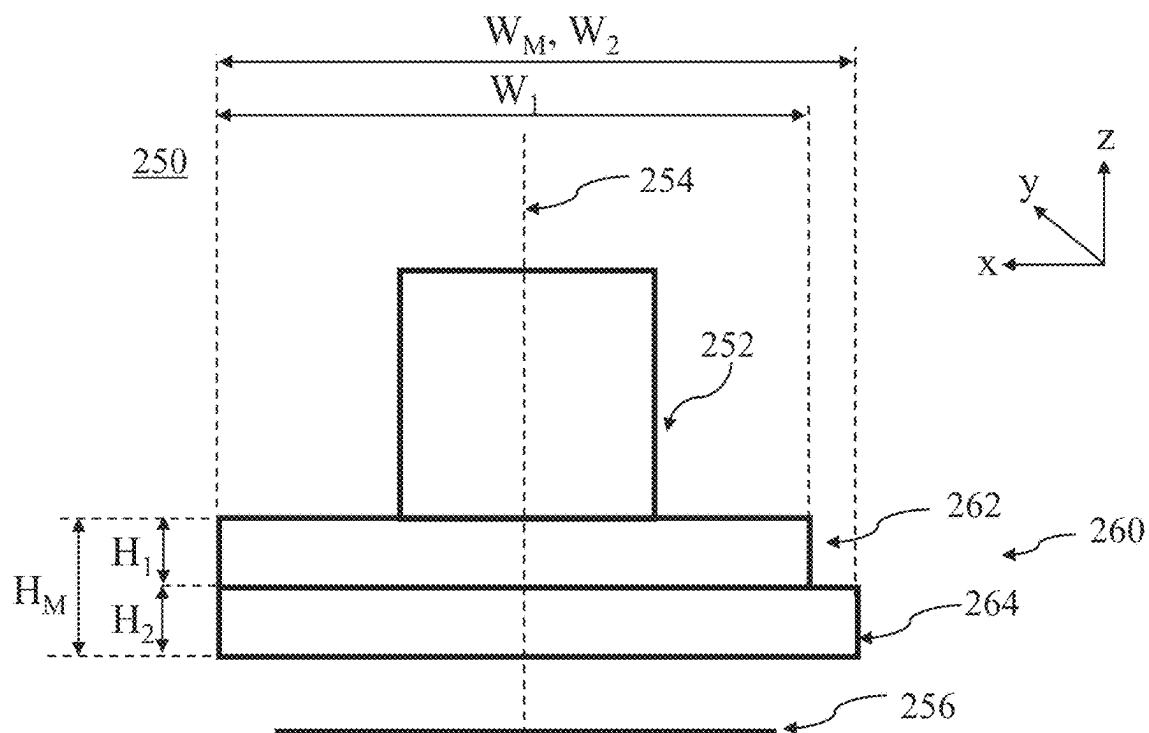
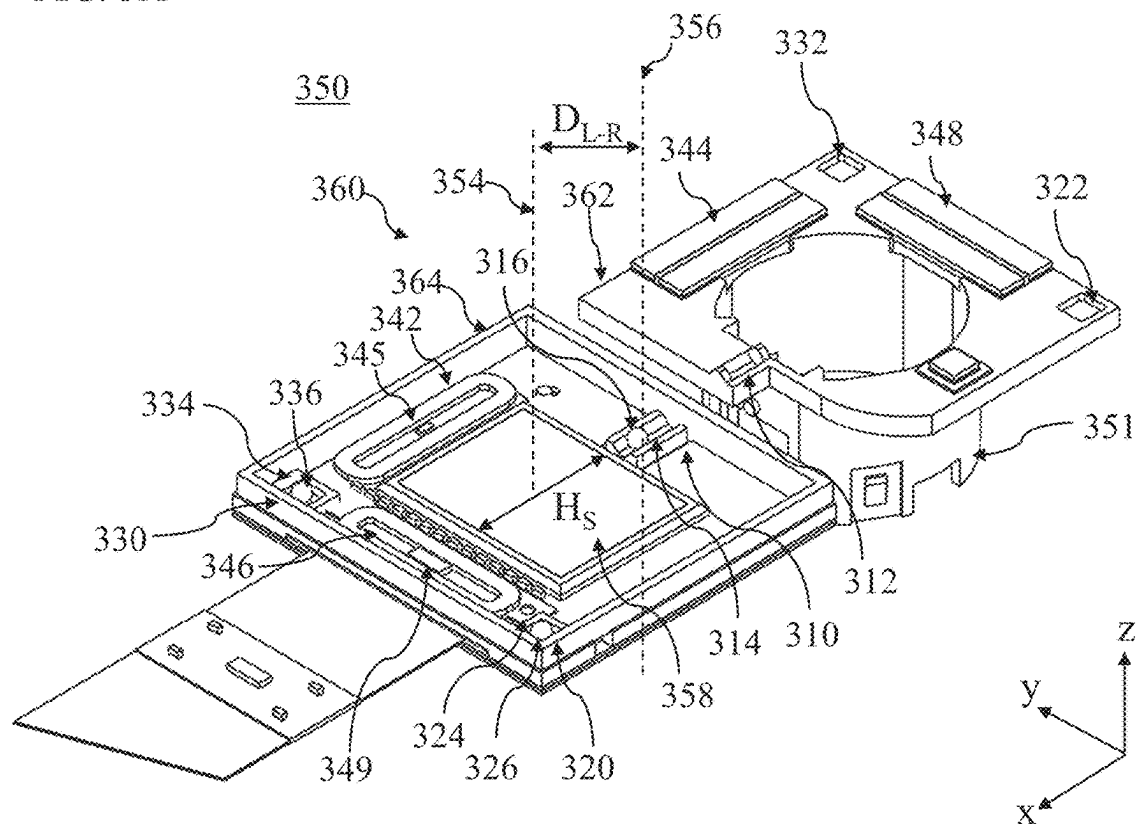
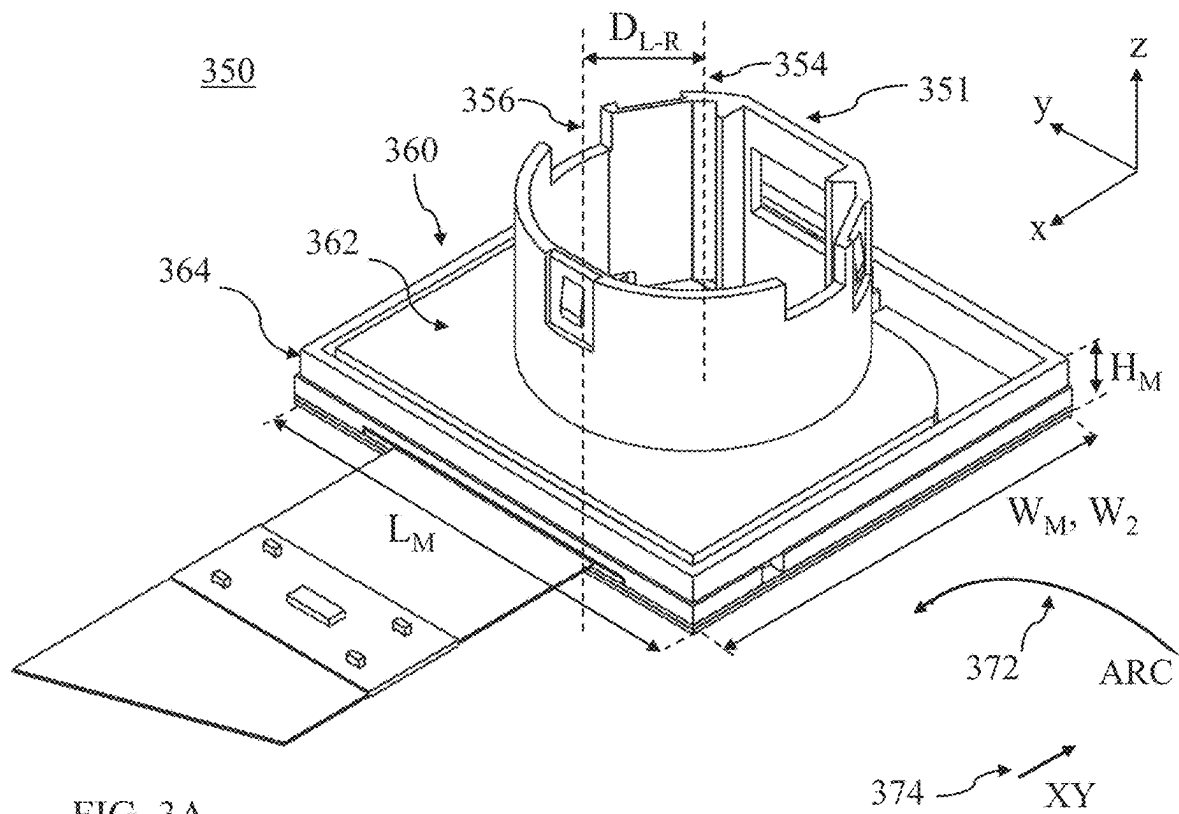
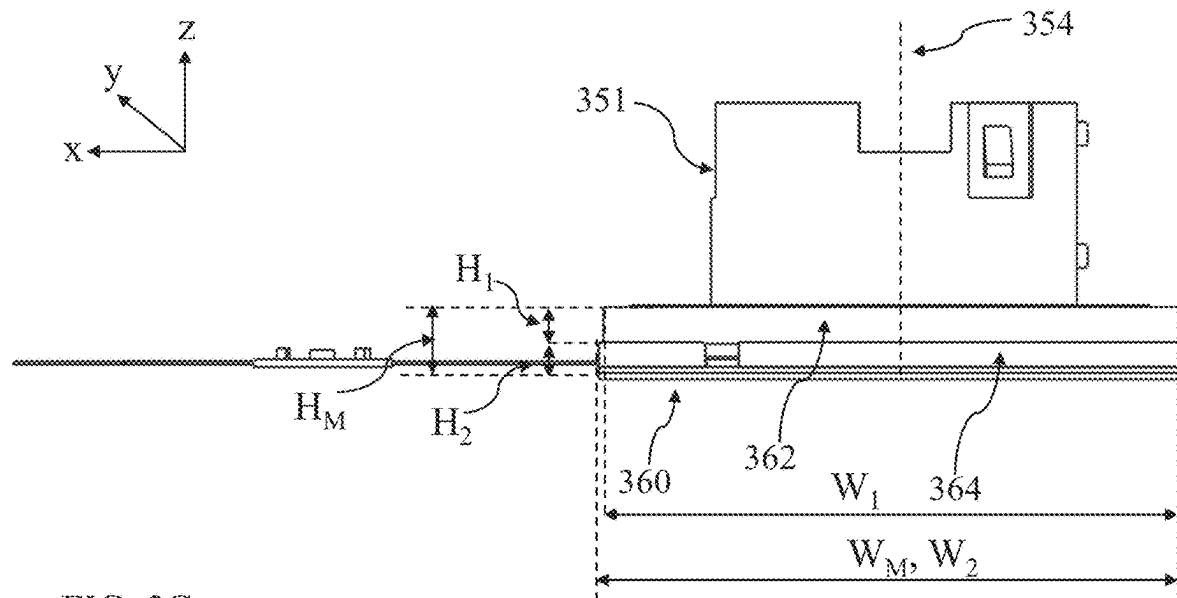


FIG. 2B





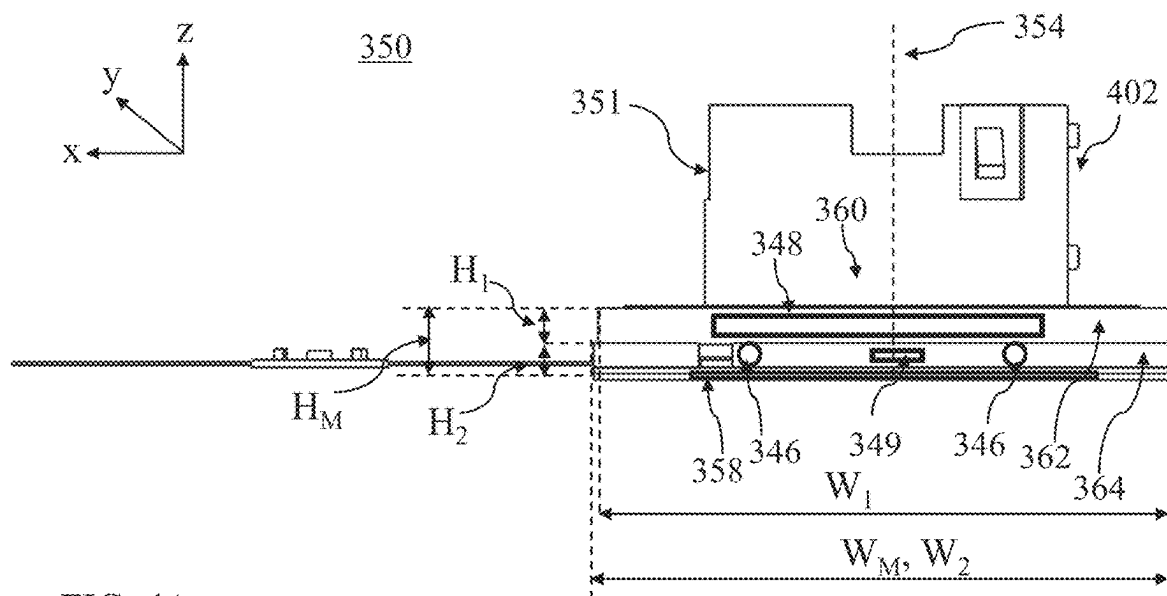


FIG. 4A

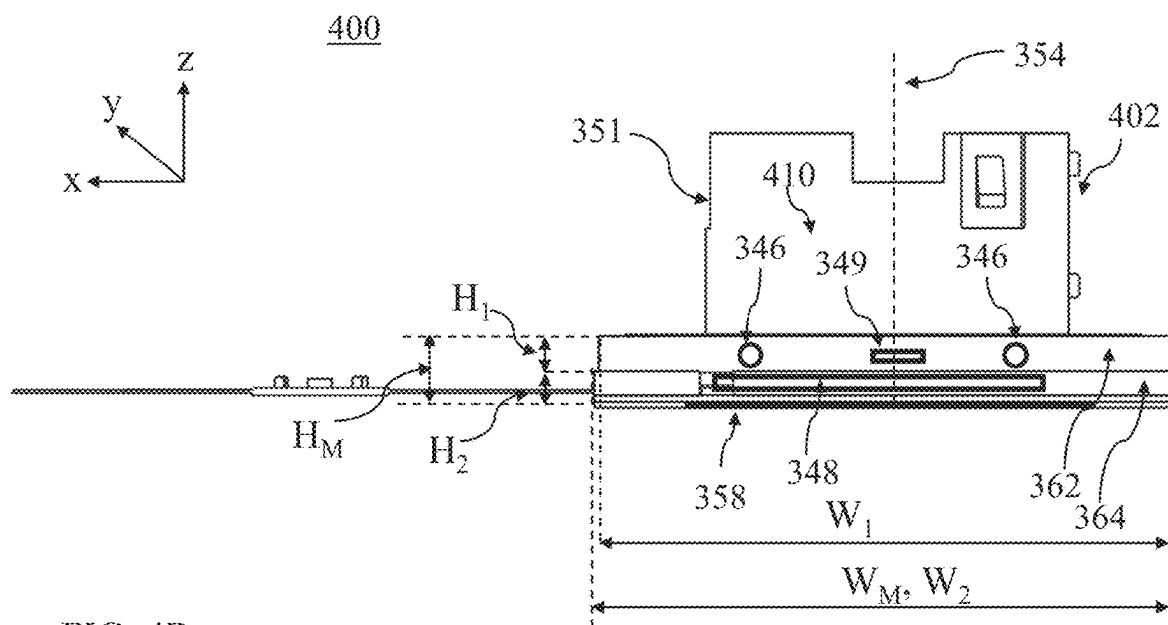


FIG. 4B

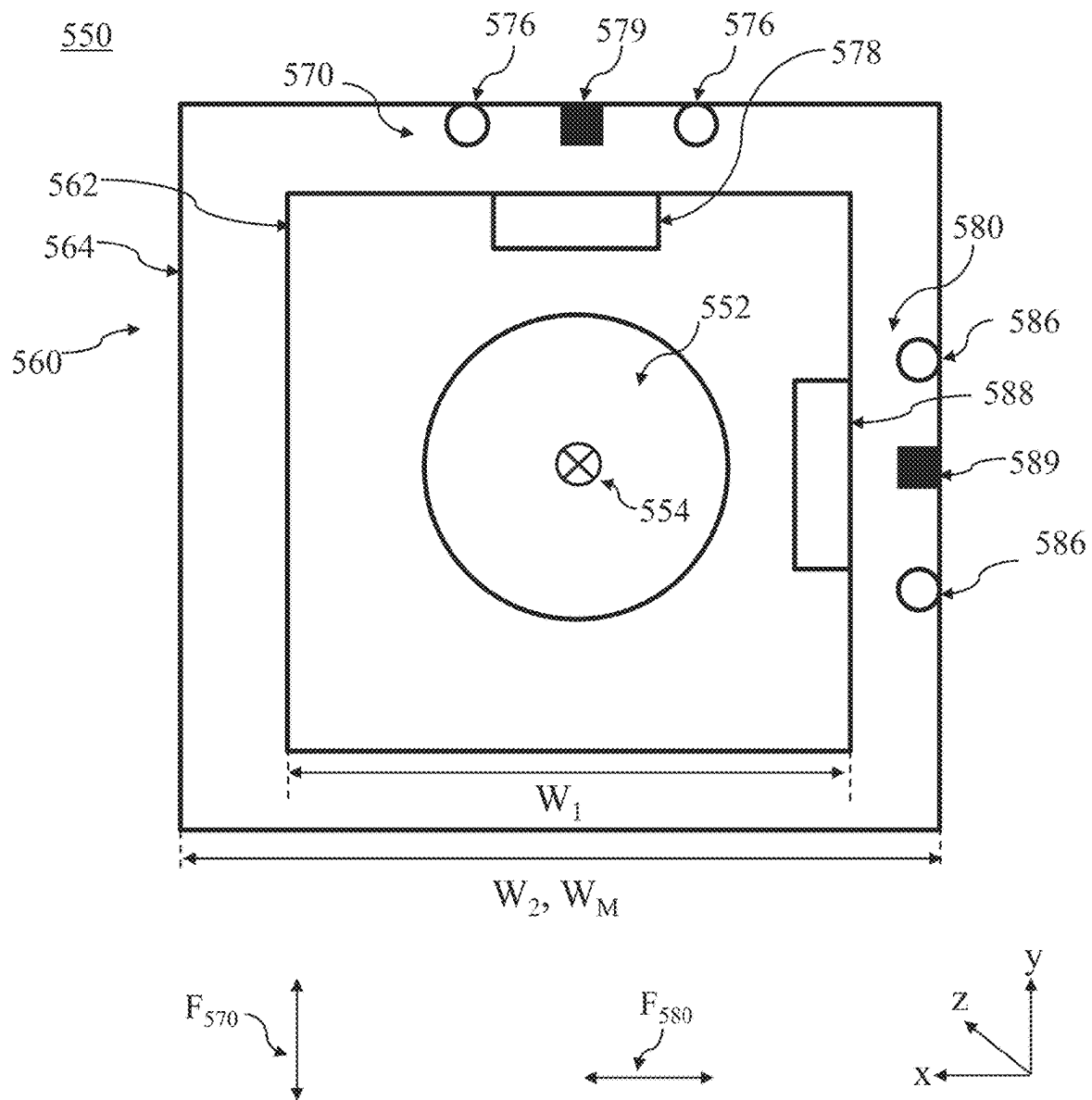


FIG. 5A

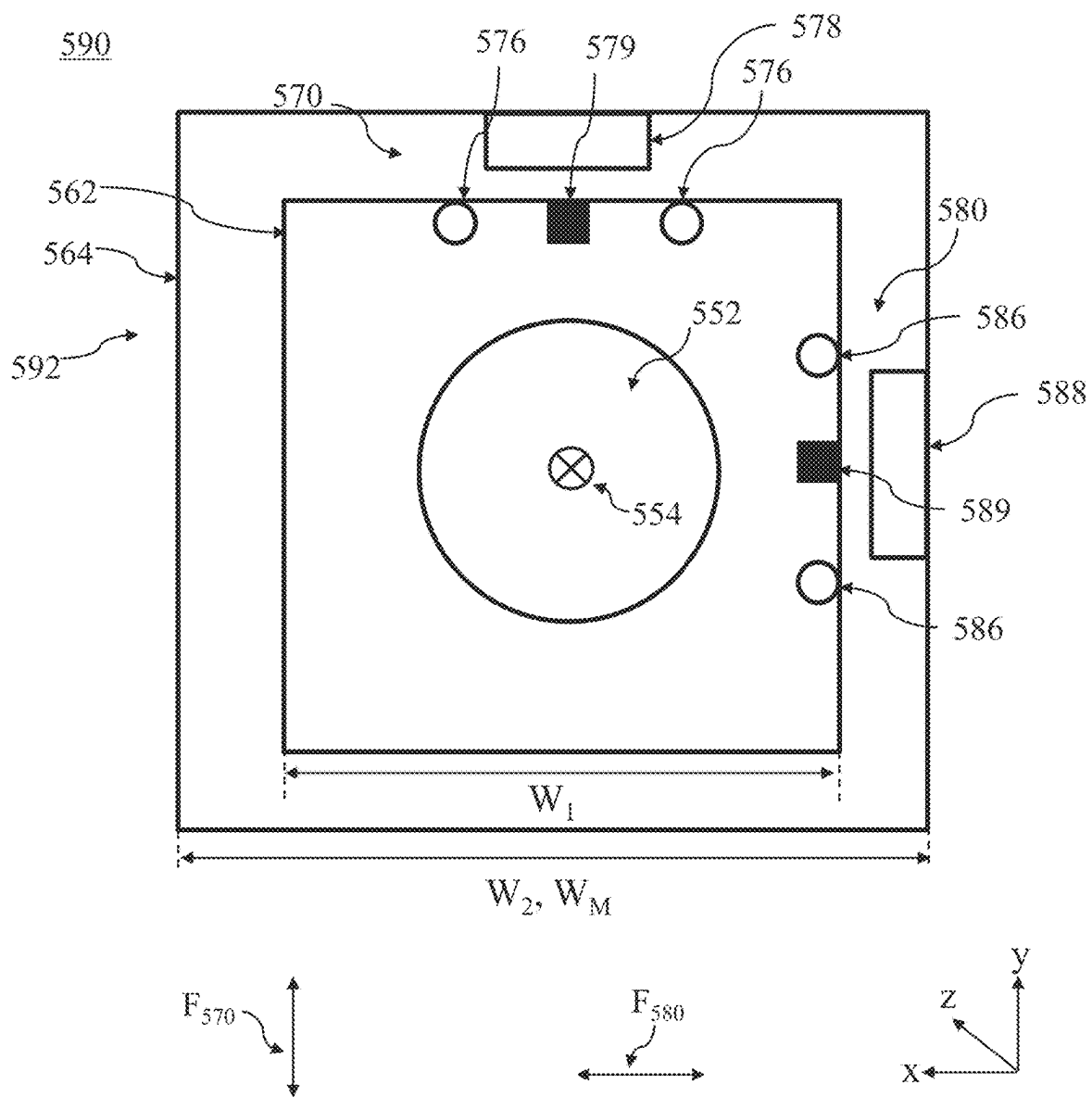


FIG. 5B

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SLIM COMPACT LENS OPTICAL IMAGE STABILIZATION

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation from U.S. patent application Ser. No. 18/847,256 filed Sep. 15, 2024 (now allowed), which was a 371 application from international patent application No. PCT/IB2023/052461 filed Mar. 14, 2023, which claims the benefit of priority from U.S. provisional patent applications Nos. 63/323,271 filed Mar. 24, 2022, 63/327,954 filed Apr. 6, 2022 and 63/408,642 filed Sep. 21, 2022, all of which are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates in general to digital cameras, and in particular to digital cameras with optical image stabilization (OIS).

BACKGROUND

Camera modules of modern mobile devices such as smartphones and tablet computers typically need to have a low thickness or height, i.e. be “slim” in order to fit into the casing of the mobile devices.

For improving a camera’s image quality, modern camera modules usually include OIS. OIS cancels (or reduces) an undesired motion of an image at the image sensor plane during the image sensor’s exposure. Without cancellation of the OIS, the undesired motion would cause a blurring of the image. Such undesired motion is for example caused by a user who unintentionally moves or shakes a mobile device during capturing of an image (“hand motion” or “hand-shake”).

In most current smartphones, OIS corrects for hand-shake around two of the three rotation axes of the device. Explicitly, OIS corrects for hand-shake around the two rotation axes which are perpendicular to a normal of an image sensor included in the camera, or in other words, perpendicular to lens optical axis of a lens included in the camera. For achieving this, movements of the lens (relative to the image sensor), of the image sensor (relative to the lens), or of the entire camera (relative to the mobile device), are performed linearly in two directions that are perpendicular to the camera optical axis. These movements are performed so that they counteract (or mitigate) a hand-shake, such that an image on the image sensor plane does not (or does only slightly) move with respect to the image sensor, i.e. the image is stabilized. If the lens is moved relative to the image sensor (and relative to the smartphone including the camera), one refers to “lens-shift OIS”.

An OIS module (or “OIS assembly”) includes all the mechanical components that are required for performing this movement. An OIS module for performing lens-shift OIS is referred to as lens-shift OIS module.

FIG. 1A shows in a top view a schematic of lens-shift OIS module for moving a lens **102** as known in the art. An image sensor (not shown) may be oriented parallel to the shown x-y plane. An optical axis (not shown) of lens **102** is oriented perpendicular to the shown x-y plane. For mitigating a user’s hand-shake (i.e. for OIS), lens **102** is moved from an initial position characterized by lens **102**’s initial center position (“ X_I ”) **104** to a final center position (“ X_F ”) **106**. For performing this movement of lens **102**, lens **102** is linearly

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moved in the x-direction as indicated by arrow **112**, and, in addition, lens **102** is linearly moved in the y-direction as indicated by arrow **114**. The two movements may be performed simultaneously.

FIG. 1B shows in a side view and schematically a camera **150** that includes a lens **152** having a lens optical axis **154**, an image sensor **156** and a lens-shift OIS module **160** as known in the art. Lens-shift OIS module **160** has an OIS module height H_M and an OIS module width W_M and is operational to perform lens-shift OIS as described in FIG. 1A. Lens-shift OIS module **160** includes a first OIS frame **162** having a first OIS frame height (“ H_1 ”) and a first OIS frame width (“ W_1 ”), a second OIS frame **164** having a second OIS frame height (“ H_2 ”) and a second OIS frame width (“ W_2 ”) and a third OIS frame **166** having a third OIS frame height (“ H_3 ”) and a third OIS frame width (“ W_3 ”). Here, $H_M = H_1 + H_2 + H_3$ and $W_M = W_3$. It is noted that a larger number of OIS frames results in a larger module height H_M .

In the x-y plane (i.e. radially with respect to optical axis **154**), lens **152** does not move relative to first OIS frame **162**. Thus, any movement in the x-y plane which is imposed on first OIS frame **162**, in particular any movement in the x-y plane for performing OIS, causes the lens to follow the movement of first OIS frame **162**. In the z-direction (i.e. axially with respect to optical axis **154**), the lens may move with respect to first OIS frame **162** and image sensor **156** for focusing. Third OIS frame **166** is static, meaning it does not move relative to image sensor **156**. Image sensor **156** does not move relative to a mobile device including camera **150**. H_1 , H_2 and H_3 may be in the range of 0.25-2.5 mm. H_M may be in the range 0.5-10 mm. W_1 , W_2 and W_3 may be in the range 5-50 mm. In particular, $W_{i+1} = W_i + 0.5-5$ mm (where $i=1, 2, 3$, etc.), i.e. given a same W_1 , a larger number of OIS frames results in a larger module width W_M .

Lens-shift OIS in a first direction perpendicular to optical axis **154** is performed by (1) linearly moving second OIS frame **164** with respect to third OIS frame **166** in a first direction, while (2) not moving first OIS frame **162** relative to second OIS frame **164**. In other words, for lens-shift OIS in the first direction, first OIS frame **162** “rides” on (or is carried by) second OIS frame **164**. Since lens **152** does not move relative to first OIS frame **162** in the x-y plane, this leads to a linear movement of lens **152** in the first direction. Lens-shift OIS in a second direction perpendicular to both the first direction and to the optical axis **154** is performed by (1) linearly moving first OIS frame **162** with respect to second OIS frame **164** in the second direction while (2) not moving second OIS frame **164** relative to third OIS frame **166**. This leads to a linear movement of lens **152** in the second direction.

In some examples, e.g. such as shown in co-owned international patent application PCT/IB2022/052194, which is incorporated herein by reference in its entirety, actuation for lens-shift OIS in a first direction is transmitted by a “ball-bearing” mechanism as known in the art. For transmitting this actuation, one or more balls of the ball-bearings are enclosed in and move in a linear rail formed in an interspace by two grooves, each groove being included in a different component (or part). An example of a ball-bearing mechanism is pin-groove mechanism **310** (FIG. 3B).

OIS actuation may for example be provided by a voice coil motor (VCM). For transmitting actuation for lens-shift OIS in the first direction, second OIS frame **164** includes one or more grooves that are oriented parallel to the first direction, and third OIS frame **166** includes one or more grooves that face the one or more grooves included in second OIS frame **164**. One or more moving balls are enclosed (or

confined) in one or more rails formed by the grooves included in second OIS frame **164** and in third OIS frame **166**. Transmission of an actuation for lens-shift OIS in the second direction may be performed in a same manner.

The motion required for OIS in the x-y plane is fully defined within a certain OIS range, i.e. within the OIS range lens **152** can be moved from any X_I to any X_F , given that X_I and X_F are included in the OIS range. We note that lens-shift OIS in the first linear direction and lens-shift OIS in the second linear direction cannot be transmitted using two OIS frames only. Therefore, three OIS frames are used, which add additional height and width (i.e. cause a height and width "penalty").

A slim and compact lens-shift OIS module, i.e. an OIS lens-shift module having a low H_M and a low W_M is beneficial for achieving a slim and compact camera module. Therefore, there is need and it would be beneficial to have a slim and compact OIS module.

SUMMARY

In various exemplary embodiments, there are provided OIS mechanisms comprising a moving frame including a first groove; a static frame including a second groove; an OIS actuator; and first, second and third ball-bearings defining an OIS plane and positioned between the moving frame and the static frame to allow a first movement and a second movement of the moving frame relative to the static frame, wherein the first ball bearing is located in a rail formed by the first groove and the second groove, wherein the first movement is a rotational movement of the moving frame around a rotation axis that coincides with the position of the first ball bearing and is perpendicular to the OIS plane, and wherein the second movement is a linear movement of the moving frame along the rail.

In some examples, the moving frame includes a first pool and a third pool, wherein the static frame includes a second pool and a fourth pool, wherein the second ball bearing is located in a volume formed by the first pool the second pool, and wherein the third ball bearing is located in a volume formed by the third pool and the fourth pool.

In some examples, the OIS mechanism is operational to provide OIS along two OIS directions.

In some examples, the OIS actuator is a regular voice coil motor (VCM). In some examples, the regular VCM includes at least one magnet fixedly coupled to the static frame. In some examples, the regular VCM includes at least one magnet fixedly coupled to the moving frame.

In some examples, the OIS actuator is a push-pull VCM. In some examples, the push-pull VCM includes at least one magnet fixedly coupled to the static frame. In some examples, the push-pull VCM includes at least one magnet fixedly coupled to the moving frame.

In some examples, the OIS mechanism is operational to provide OIS in an OIS range of less than 2 mm in the direction of the first movement and less than 2 mm in the direction of the second movement.

In some examples, the moving frame has a moving frame height H_{MF} and the static frame has a static frame height H_{SF} , wherein H_{MF} is in the range of 0.25 mm to 1.5 mm and wherein H_{SF} is in the range of 0.25 mm to 1.5 mm.

In some examples, the moving frame has a moving frame width W_{MF} and the static frame has a static frame width W_{SF} , wherein W_{MF} is in the range of 10 mm to 40 mm and wherein W_{SF} is in the range of 10 mm to 40 mm.

In some examples, the OIS mechanism is included in an OIS module, and wherein the OIS module has an OIS module height H_M in the range of 0.5 mm to 5 mm.

In some examples, the OIS mechanism is included in an OIS module, and wherein the OIS module has an OIS module width W_M in the range of 10 mm to 40 mm.

In some examples, an OIS mechanism is included in a digital camera that comprises a lens, and the lens moves together with the moving frame for the OIS. In some examples, the lens moves relative to the moving frame for focusing. In some examples, the digital camera includes an image sensor and the position of the first ball bearing is located at a center position of the image sensor. In some examples, the digital camera is a pop-out camera. In some examples, the digital camera is included in a smartphone.

In various exemplary embodiments, there is provided, in a mobile device comprising a digital camera that includes a lens and an image sensor, a method comprising: providing in the digital camera a lens-shift OIS module that includes only a first, moving frame, and a second, static frame; and actuating the moving frame to move the lens relative to the image sensor in a first, rotational movement, and in a second, linear movement to provide OIS, wherein the lens has an effective focal length (EFL) in the range of 2.5 mm to 50 mm.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A shows in a top view a schematic of a known art lens-shift OIS (optical image stabilization) moving a lens;

FIG. 1B shows in a side view and schematically a camera that includes a lens having a, an image sensor and a lens-shift OIS module as known in the art;

FIG. 2A shows a schematic of a lens-shift OIS moving a lens as disclosed herein in a top view;

FIG. 2B shows schematically a camera including a lens, an image sensor and a lens-shift OIS module as disclosed herein in a side view;

FIG. 3A shows parts of a camera module that includes a lens-shift OIS module as disclosed herein in a perspective view;

FIG. 3B shows parts of the camera module of FIG. 3A in another perspective view;

FIG. 3C shows parts of the camera module of FIGS. 3A-B in a side view;

FIG. 4A shows parts of the camera module of FIGS. 3A-B in another side view;

FIG. 4B shows parts of another camera module disclosed herein in the same side view as shown in FIG. 4A;

FIG. 5A shows schematically a camera including a lens and another lens-shift OIS module as disclosed herein in a top view;

FIG. 5B shows schematically a camera including a lens and a yet another lens-shift OIS module as disclosed herein in a top view.

DETAILED DESCRIPTION

FIG. 2A shows a schematic of a lens-shift OIS moving a lens **202** as disclosed herein in a top view. An image sensor (not shown) may be oriented parallel to the shown x-y plane. An optical axis (not shown) of lens **202** is oriented perpendicular to the shown x-y plane. For OIS, lens **202** is moved from an initial position characterized by lens **202**'s initial center position (" X_I ") **204** to its final center position (" X_F ") **206**. For performing that movement of lens **202**, lens **202** may be rotationally moved around an axis (FIGS. 3A-B)

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substantially parallel to the optical axis of lens **202** and as indicated by an arc **212**, and, in addition, lens **202** may be linearly moved in the y-direction as indicated by an arrow **214**.

FIG. 2B shows schematically an embodiment of a camera numbered **250** that includes a lens **252** having a lens optical axis **254**, an image sensor **256** and a lens-shift OIS module **260** as disclosed herein in a side view. Lens-shift OIS module **260** has an OIS module height H_M , an OIS module width W_M and is operational to perform lens-shift OIS as described in FIG. 2A. Lens-shift OIS module **260** includes a first OIS frame **262** having a first OIS frame height H_1 and a first OIS frame width W_1 and a second OIS frame **264** having a second OIS frame height H_2 and a second OIS frame width W_2 . Here, $H_M = H_1 + H_2$ and $W_M = W_2$. In comparison with known OIS module **160**, in OIS module **260** there is no need for a third OIS frame, and therefore an additional height penalty and width penalty associated with a third OIS frame is avoided. This allows a slim and compact OIS module, i.e. an OIS module having a low H_M and a small W_M , which is beneficial for use in a camera included in a mobile device such as a smartphone. To clarify, all embodiments disclosed herein are beneficial for use in a camera included in a mobile device such as a smartphone. The camera may include a lens having a lens optical axis, an effective focal length ("EFL") in the range of 2.5 mm-50 mm, and preferably in the range of 5 mm-20 mm. The camera may as well include an image sensor having an image sensor diagonal ("SD") in the range of in the range of 5 mm-30 mm, and preferably in the range of 7.5 mm-25 mm. In some examples, such a camera may be a sub-camera which, together with other sub-cameras forms a multi-camera (such as a dual-camera) of a mobile device, as known in the art.

In the x-y plane, lens **252** does not move relative to first OIS frame **262**. Thus, any movement in the x-y plane imposed on first OIS frame **262**, in particular any movement in the x-y plane for performing OIS, causes lens **252** to follow the movement of first OIS frame **262**. Therefore, first OIS frame **262** is also referred to as "moving frame". In the z-direction, lens **252** may move with respect to first OIS frame **262** and image sensor **256** for focusing. Second OIS frame **264** is static and does not move relative to image sensor **256**. Therefore, second OIS frame **264** is also referred to as "static frame". Image sensor **256** does not move relative to a mobile device including camera **250**. This is valid for all embodiments disclosed herein. In the following and when referring to an OIS component that does not move (or is static) relative to a mobile device including an OIS mechanism disclosed herein, we state that "the OIS component does not move with respect to the image sensor". W_M is determined by W_2 , i.e. $W_M = W_2$. In particular, $W_{i+1} = W_i + 0.5-5$ mm, i.e. given a same W_1 , a smaller number of OIS frames results in a smaller module width W_M , which is beneficial for use in a camera included in a mobile device such as a smartphone. In some embodiments, lens-shift OIS modules disclosed herein may be beneficial for use in a "pop-out camera", such as e.g. disclosed in the international patent application PCT/IB2020/058697, which is incorporated herein by reference in its entirety.

Lens-shift OIS in the x-y plane, i.e. perpendicular to the optical axis **254**, is performed by (1) rotationally moving first OIS frame **262** with respect to second OIS frame **264** in a rotation direction as indicated by arc **212** while (2) in addition linearly moving first OIS frame **262** relative to second OIS frame **264** as indicated by arrow **214**. Since lens **252** does not move relative to first OIS frame **262** in the x-y

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plane, this leads to a superposition of a rotational and a linear movement of lens **252** in the x-y plane. It is noted that these two movements can be performed sequentially or simultaneously. An example for actuation and transmission of this actuation for lens-shift OIS is described in FIGS. 3A-C, FIGS. 4A-B and FIGS. 5A-B.

The motion required for OIS within a certain OIS range in the x-y plane is fully defined, i.e. within the OIS range lens **252** can be moved from any X_I to any X_F , given that X_I and X_F are included in the OIS range. We note that lens-shift OIS in the entire OIS range in the x-y plane can be performed using two OIS frames. Therefore, a height and width penalty associated with a third OIS frames is prevented. Typically, an OIS range covers an area of less than 10 mm×10 mm. Often, it may cover an area of less than 2 mm×2 mm or even less than 1 mm×1 mm.

FIG. 3A shows parts of an embodiment of a camera module numbered **350** which includes a lens-shift OIS module **360** as disclosed herein in a perspective view. An image sensor (not visible) may be oriented parallel to the shown x-y plane. Camera module **350** includes a lens carrier **351** that includes a lens (not shown) having a lens optical axis **354**. Lens optical axis **354** is oriented perpendicular to the x-y plane, i.e. parallel to the z-axis. In the x-y plane, the lens does not move relative to lens carrier **351**, so that an x-y movement imposed on lens carrier **351** causes the lens to follow the movement of lens carrier **351**. In particular, any movement imposed on lens carrier **351** for OIS is performed also by the lens.

Lens-shift OIS module **360** has an OIS module height H_M , an OIS module width W_M and an OIS module length L_M . Lens-shift OIS module **360** includes a first OIS frame **362** having a first OIS frame height (H_1 , see FIG. 3C) and a first OIS frame width (W_1 , see FIG. 3C) and a second OIS frame **364** having a second OIS frame height (H_2 , see FIG. 3C) and a second OIS frame width (W_2 , see FIG. 3C). H_1 and H_2 may be in the range $H_1, H_2 = 0.1-5$ mm, preferably H_1 and H_2 may be in the range $H_1, H_2 = 0.25-1.5$ mm. H_M is determined by H_1 and H_2 , i.e. $H_M = H_1 + H_2$. H_M may be in the range $H_M = 0.25-10$ mm, preferably H_M may be in the range $H_M = 0.5-5$ mm. W_M and L_M are to a large extent defined by a size, i.e. a width and a height (such as H_S), of image sensor **358**. $L_M = W_M + 50\%$. W_1 and W_2 may be in the range $W_1, W_2 = 5-75$ mm, preferably W_1 and W_2 may be in the range $W_1, W_2 = 10-40$ mm. W_M is determined by W_2 , i.e. $W_M = W_2$. In particular, $W_2 = W_1 + 0.5-5$ mm, i.e. given a same W_1 , a smaller number of OIS frames results in a smaller module width W_M , which is beneficial for use in a camera included in a mobile device such as a smartphone. External dimensions of OIS module **360** in the x-y plane may be such that OIS module **360** may fit in a circle (i.e. may be encircled by a circle) having a diameter between 5 and 75 mm. Lens-shift OIS module **360** includes a linear pin-groove mechanism **310**, a first pin-pool mechanism **320**, a second pin-pool mechanism **330**, a first OIS VCM and a second OIS VCM as described below. The first OIS VCM and the second OIS VCM act as OIS actuator, i.e. the role (or function) of the first OIS VCM and the second OIS VCM in lens-shift OIS module **360** is to actuate an OIS motion in two directions. For two-directional (or two-dimensional) OIS motion, the first OIS VCM actuates an OIS motion in a first direction and the second OIS VCM actuates an OIS motion in a second direction. The role of linear pin-groove mechanism **310**, first pin-pool mechanism **320** and second pin-pool mechanism **330** is to transmit and direct the OIS motion.

Lens carrier **351** is fixedly coupled to first OIS frame **362**. First OIS frame **362** is a moving frame. Second OIS frame

364 is a static frame and does not move relative to image sensor 358 (see FIG. 3B). For OIS, first OIS frame 362 is rotationally moved relative to second OIS frame 364 as indicated by arc 372. The rotational movement is around a rotation axis 356 which is substantially parallel to the lens optical axis 354. A distance between lens optical axis 354 and rotation axis 356 is marked " D_{L-R} ". D_{L-R} may be in the range of $D_{L-R} = \frac{1}{2} H_S + 0 \text{ mm} - 10 \text{ mm}$, wherein H_S is a height of image sensor 358. There is a trade-off for a size of D_{L-R} : A large D_{L-R} is beneficial for performing a rotational movement having a large radius of curvature, because a large radius of curvature more resembles a linear movement and which is beneficial for simple actuation control. On the other hand, a small D_{L-R} is beneficial for a compact lens-shift OIS module, i.e. a lens-shift OIS module having a small W_M . In addition, first OIS frame 362 is linearly moved in the x-y plane as indicated by arrow 374.

FIG. 3B shows parts of camera module 350 of FIG. 3A in another perspective view. FIG. 3C shows parts of camera module 350 of FIGS. 3A-B in a side view. Camera module 350 includes an image sensor 358, a linear pin-groove mechanism 310 ("first ball-bearing"), a first pin-pool mechanism 320 ("second ball-bearing") and a second pin-pool mechanism 330 ("third ball-bearing") for performing the rotational OIS motion and the linear OIS motion described above and as indicated by arc 372 and by arrow 374 respectively. Included in a mobile device, lens-shift OIS module 360 provides OIS along two directions, i.e. it corrects for hand-shake around two rotation axes. Specifically, lens-shift OIS module 360 corrects for hand-shake around the two rotation axes which are perpendicular to a normal of image sensor 358, or in other words, perpendicular to lens optical axis 354. This is valid for all lens-shift OIS modules disclosed herein. The OIS is provided by moving two frames relative to each other, wherein the relative movement is a rotational movement and a linear movement.

Linear pin-groove mechanism 310 includes a first groove 312 that is included in first OIS frame 362, a second groove 314 that is included in second OIS frame 364, and a ball of a ball-bearing (or "bearing ball" or simply "ball") 316. First OIS VCM includes a coil 346, a magnet 348 and a position sensor 349 (e.g. a magnetic flux measuring device ("MFMD") such as a Hall sensor) and actuates the linear motion of first OIS frame 362 relative to second OIS frame 364. Magnet 348 is fixedly coupled to first OIS frame 362. Coil 346 and position sensor 349 are fixedly coupled to second OIS frame 364. The linear motion is transmitted via linear pin-groove mechanism 310, i.e. by ball 316 which is enclosed in a rail formed by first groove 312 and second groove 314. First pin-pool mechanism 320 includes a first pool 322 (or "recess" or "notch") that is included in first OIS frame 362, a second pool 324 that is included in second OIS frame 364 and a ball 326. Second pin-pool mechanism 330 includes a first pool 332 that is included in first OIS frame 362, a second pool 334 that is included in second OIS frame 364, and a ball 336. Second OIS VCM includes a coil 342, a magnet 344 and a position sensor 345 (e.g. a Hall sensor) and actuates the rotational motion of first OIS frame 362 around rotation axis 356 relative to second OIS frame 364. Magnet 344 is fixedly coupled to first OIS frame 362. Coil 342 and position sensor 345 are fixedly coupled to second OIS frame 364. The rotational motion is transmitted via first pin-pool mechanism 320 and second pin-pool mechanism 330, i.e. ball 326 and ball 336 which are enclosed in a volume (or space) formed by first pool 322 and second pool 324 and first pool 332 and second pool 334 respectively.

Rotation axis 356 coincides with the position of ball 316, i.e. ball 316 forms a pivot point for rotation.

In lens-shift OIS module 360 and with respect to the y-axis, the pivot point is located at a center position of image sensor 358. The center position defines a symmetry axis of image sensor 358. In other embodiments and with respect to the y-axis, linear pin-groove mechanism 310 (and the pivot point) may be located at another location. That is, the pivot point may not be located at a symmetry axis of image sensor 358. In other embodiments, springs as known in the art instead of the ball-bearings may transmit and direct the OIS motion.

FIG. 4A shows parts of camera module 350 of FIGS. 3A-C in another side view. All components and dimensions are identical to the parts of a camera module 350 shown in FIG. 3C. In particular, the parts of a camera module 350 include lens-shift OIS module 360 as described above. First OIS VCM including coil 346, magnet 348 and position sensor 349 are visible. A VCM actuator for focusing a lens ("focus actuator") included in lens carrier 351 may include a coil, a position sensor and a magnet. The coil and the position sensor of the focus actuator may be fixedly coupled to lens carrier 351, a position (or location) of the coil and the position sensor is marked 402. In particular, this means that there is an electrical connection between a mobile device including the parts of a camera module 350 and lens carrier 351. The electrical connection is operational to supply power and control signals (and receive position signals) to the coil and to the position sensor of the focus actuator.

FIG. 4B shows parts of another embodiment of a camera module numbered 400 disclosed herein in the same side view as shown in FIG. 4A. Camera module 400 may be identical with camera module 350 in terms of functionality and dimensions. All components included in the parts of camera module 400 as well as the dimensions (W_1 , W_M , W_2 etc.) of the parts of camera module 400 respectively are identical to the components included in the parts of camera module 350 as well as the dimensions of the parts of camera module 350 respectively. However, the parts of camera module 400 include another lens-shift OIS module 410. Lens-shift OIS module 410 differs from lens-shift OIS module 360 (see FIG. 4A) by the location (or position) of components included in the respective OIS actuator. The OIS actuator in lens-shift OIS module 410 includes a third VCM and a fourth VCM. All components of the first and second OIS VCMs in lens-shift OIS module 360 are also included in the third and fourth OIS VCMs of lens-shift OIS module 410. However, the inclusion of the components is inverse: all components of the first OIS VCM and the second OIS VCM respectively included in second OIS frame 364 in the third OIS VCM and the fourth OIS VCM respectively, are here included in first OIS frame 362 and vice versa. Explicitly, this means that magnet 348 is fixedly coupled to second OIS frame 364, and coil 346 and position sensor 349 are fixedly coupled to first OIS frame 362, as shown. Magnet 344 is fixedly coupled to second OIS frame 364, and coil 342 and position sensor 345 are fixedly coupled to first OIS frame 362 (not shown). In camera module 350, magnet 348 is farther from image sensor 358 than position sensor 349. In camera module 400, magnet 348 is closer to image sensor 358 than position sensor 349.

In some examples, an electrical connection to coil 342 and position sensor 345 as well as to coil 346 and position sensor 349 may be provided by the very same electrical connection between a mobile device including camera module 400 and lens carrier 351 (which is fixedly coupled to second OIS frame 364) as described above.

For controlled motion, a VCM performs position sensing as well as actuation. Some advantages of lens-shift OIS module 410 over lens-shift OIS module 360 may include (1) a simpler position sensing, (2) a simpler actuation control, and/or (3) a faster actuation, as detailed below. It is noted that magnet 344 and magnet 348 respectively are used both for OIS actuation (due to interaction with coil 342 and coil 346 respectively) and for position sensing (due to interaction with position sensor 345 and position sensor 349 respectively), whereas position sensor 345 and position sensor 349 are used for position sensing only.

(1) Simpler Position Sensing

A symmetry axis for position sensing is defined by a relative orientation between magnet 344 and image sensor 358 and between magnet 348 and image sensor 358 respectively. When performing OIS with lens-shift OIS module 360, both magnet 344 and magnet 348 move rotationally with respect to image sensor 358. This means that a symmetry axis for position sensing rotates with respect to image sensor 358. When performing OIS with lens-shift OIS module 410, both magnet 344 and magnet 348 do not move with respect to image sensor 358. This means that a symmetry axis for position sensing does not rotate with respect to image sensor 358, which is beneficial for sensing OIS movements.

(2) Simpler Actuation Control

A symmetry axis of magnet 344 and magnet 348 respectively determines a direction of the respective VCM's actuation force. When performing OIS with lens-shift OIS module 410, both magnet 344 and magnet 348 do not move with respect to image sensor 358. This means that a direction of the VCM's actuation force does not rotate with respect to image sensor 358, which is beneficial for controlling OIS actuation.

(3) Faster Actuation

In lens-shift OIS module 410, magnet 344 and magnet 348 respectively do not move with respect to image sensor 358, i.e. magnet 344 and magnet 348 respectively are not actuated for performing OIS. In other words, magnet 344 and magnet 348 are not included in the "moving mass" of OIS module 410. In general, a magnet makes up a relatively large share (or portion) of a weight of a VCM. This means that compared to lens-shift OIS module 360, lens-shift OIS module 410, has a relatively small moving mass, what is beneficial in terms of fast (or low-power) actuation.

FIG. 5A shows schematically an embodiment of another camera module numbered 550 that includes a lens 552 having a lens optical axis 554 and a lens-shift OIS module 560 as disclosed herein. Lens-shift OIS module 560 includes a first OIS frame 562 having a first OIS frame width W_1 and a second OIS frame 564 having a second OIS frame width W_2 . OIS module width W_M is determined by second OIS frame 564, i.e. $W_M = W_2$. All details of the movement of first OIS frame 562 and second OIS frame 564 as well as lens 552 are identical as described above for OIS module 260, OIS module 360, etc.

Camera module 550 may be identical with camera module 350 and camera module 400 in terms of functionality and dimensions. Lens-shift OIS module 560 may be identical with lens-shift OIS module 360 or it may be identical with lens-shift OIS module 410 in terms of OIS functionality and dimensions, except that lens-shift OIS module 560 includes a different OIS actuator. The OIS actuators included in lens-shift OIS module 360 and in lens-shift OIS module 410 are characterized by the fact that during an operation of a VCM included in the OIS actuator, a distance between a coil (or a position sensor such as a magnetic flux measuring

device ("MFMD") included in the VCM and a magnet included in the VCM does not change. In other words, a first plane oriented parallel to the coil and a second plane oriented parallel to the magnet's movement are parallel to each other and, in addition, a distance between the first plane and the second plane does not change. In the following, we refer to these VCMs as "regular VCM". The OIS actuators included in lens-shift OIS module 360 and in lens-shift OIS module 410 are regular VCMs. In contrast, lens-shift OIS module 560's OIS actuator includes a different type of VCM, which we refer to as "push-pull VCM". Specifically, lens-shift OIS module 560 includes a first push-pull VCM 570 and a second push-pull VCM 580. A direction of a force that is exercised (or generated) by first push-pull VCM 570 and second push-pull VCM 580 respectively is indicated by arrow F_{570} (parallel to the y-axis) and F_{580} (parallel to the x-axis) respectively. A force of first push-pull VCM 570 and second push-pull VCM 580 respectively is transmitted into an OIS movement as detailed above, e.g. based on ball-bearings. First push-pull VCM 570 includes a coil 576, a magnet 578 and a position sensor 579. Second push-pull VCM 580 includes a coil 586, a magnet 588 and a position sensor 589. As shown, magnet 578 and magnet 588 are fixedly coupled to first OIS frame 562. First OIS frame 562 is a moving frame. Coil 576 and position sensor 579 as well as coil 576 and position sensor 579 are fixedly coupled to second OIS frame 564. Second OIS frame 562 is a static frame.

FIG. 5B shows schematically an embodiment of another camera module numbered 590 that includes a lens-shift OIS module 592 as disclosed herein. Camera module 590 may be identical with camera module 350, camera module 400 and camera module 55 in terms of functionality and dimensions. The OIS actuator of lens-shift OIS module 592 includes a third push-pull VCM and a fourth push-pull VCM. All components of lens-shift OIS module 560's first push-pull VCM and second push-pull VCM are included in lens-shift OIS module 592's third push-pull VCM and fourth push-pull VCM as well. However, the inclusion of the components is inverse: in camera module 590, magnet 578 and magnet 588 are fixedly coupled to second OIS frame 564, and coil 576 and position sensor 579 as well as coil 576 and position sensor 579 are fixedly coupled to first OIS frame 562. The advantages of such an embodiment are identical to advantages of lens-shift OIS module 410 over lens-shift OIS module 360 as discussed above.

For the sake of clarity, the term "substantially" is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term "substantially" used herein should be interpreted to imply possible variation of up to 10% over or under any specified value. According to another example, the term "substantially" used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to a further example, the term "substantially" used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value.

It is to be noted that the various features described in the various embodiments can be combined according to all possible technical combinations.

It is to be understood that the disclosure is not limited in its application to the details set forth in the description contained herein or illustrated in the drawings. The disclosure is capable of other embodiments and of being practiced and carried out in various ways. Hence, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as

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limiting. As such, those skilled in the art will appreciate that the conception upon which this disclosure is based can readily be utilized as a basis for designing other structures, methods, and systems for carrying out the several purposes of the presently disclosed subject matter.

Those skilled in the art will readily appreciate that various modifications and changes can be applied to the embodiments of the disclosure as hereinbefore described without departing from its scope, defined in and by the appended claims.

What is claimed is:

1. An optical image stabilization (OIS) mechanism, comprising:

a lens having a lens optical axis and an effective focal length EFL in the range of 2.5 mm to 50 mm;

an image sensor having a sensor height H_S smaller than a sensor width;

a magnet;

a moving frame; and

a static frame fixedly coupled to the image sensor, wherein the OIS mechanism allows a first movement and a second movement of the moving frame relative to the static frame,

wherein the first movement is a rotational movement of the moving frame around a rotation axis parallel to the lens optical axis,

wherein a distance between the rotation axis and the lens optical axis is marked D_{L-R} ,

wherein D_{L-R} is in the range of $\frac{1}{2} H_S$ to $\frac{1}{2} H_S + 10$ mm,

wherein the second movement is a linear movement, wherein the lens moves together with the moving frame for OIS,

wherein the magnet is used for actuating one of the first movement or the second movement and for sensing a position of the lens relative to the image sensor, and wherein the OIS mechanism is included in a mobile device.

2. The OIS mechanism of claim 1, wherein the moving frame includes a first groove and the static frame includes a second groove, wherein the OIS mechanism includes a first, a second and a third ball-bearing, and wherein the three ball-bearings define a plane in which the first movement and a second movement occur.

3. The OIS mechanism of claim 2, wherein the first ball bearing is located in a rail formed by the first groove and the second groove, and wherein the rotation axis coincides with a position of the first ball bearing.

4. The OIS mechanism of claim 3, wherein the linear second movement of the moving frame is along the rail.

5. The OIS mechanism of claim 2, wherein the moving frame includes a first pool and a third pool, wherein the static frame includes a second pool and a fourth pool, wherein the second ball bearing is located in a volume formed by the first pool and the second pool, and wherein the third ball bearing is located in a volume formed by the third pool and the fourth pool.

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6. The OIS mechanism of claim 1, wherein the sensing of the position of the lens relative to the image sensor is performed around a symmetry axis, wherein the OIS mechanism includes a regular voice coil motor, wherein the magnet is fixedly coupled to the static frame, and wherein the symmetry axis does not rotate with respect to the image sensor.

7. The OIS mechanism of claim 1, wherein the OIS mechanism includes a regular voice coil motor (VCM), and wherein the magnet is fixedly coupled to the moving frame.

8. The OIS mechanism of claim 1, wherein the sensing of the position of the lens relative to the image sensor is performed around a symmetry axis, wherein the OIS mechanism includes a push-pull voice coil motor, wherein the magnet is fixedly coupled to the static frame, and wherein the symmetry axis does not rotate with respect to the image sensor.

9. The OIS mechanism of claim 1, wherein the OIS mechanism includes a push-pull voice coil motor, and wherein the magnet is fixedly coupled to the moving frame.

10. The OIS mechanism of claim 1, wherein the OIS mechanism is operational to provide OIS in an OIS range of less than 2 mm in the direction of the first movement and less than 2 mm in the direction of the second movement.

11. The OIS mechanism of claim 1, wherein the moving frame has a moving frame height H_{MF} and the static frame has a static frame height H_{SF} , wherein H_{MF} is in the range of 0.25 mm to 1.5 mm and wherein H_{SF} is in the range of 0.25 mm to 1.5 mm.

12. The OIS mechanism of claim 1, wherein the moving frame has a moving frame width W_{MF} and the static frame has a static frame width W_{SF} , wherein W_{MF} is in the range of 10 mm to 40 mm and wherein W_{SF} is in the range of 10 mm to 40 mm.

13. The OIS mechanism of claim 1, wherein the OIS mechanism is included in an OIS module that has an OIS module height H_M in the range of 0.5 mm to 5 mm.

14. The OIS mechanism of claim 1, wherein the OIS mechanism is included in an OIS module that has an OIS module width W_M in the range of 10 mm to 40 mm.

15. The OIS mechanism of claim 1, wherein the lens is movable relative to the moving frame for focusing.

16. The OIS mechanism of claim 1, wherein the position of the first ball bearing is located at a center position of the image sensor.

17. The OIS mechanism of claim 1, wherein the EFL is in the range of 5 mm-20 mm.

18. The OIS mechanism of claim 1, wherein the EFL is in the range of 5 mm-15 mm.

19. The OIS mechanism of claim 1, wherein the OIS mechanism is included in a pop-out camera.

20. The OIS mechanism of claim 1, wherein the mobile device is a smartphone.

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